

Chapter 2

HyperText Markup Language (HTML)

INT150 Web Technologies (2/2025)

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Hypertext Markup Language (HTML)

- [HTML](#) is used to create most web pages.
- The development of [HTML](#) is overseen by an organization called [W3C](#) (World Wide Web Consortium).
- There have been several versions of [HTML](#).
- The last major version of [HTML](#) was [HTML 5.2](#) in 2017.
- In 2000, some stricter rules were added to [HTML 4.01](#). It is known as [XHTML](#) ([Extensible Hypertext Markup Language](#))

XHTML

- is semantically identical to [HTML 4.01](#) but is stricter than HTML.
- The primary syntactic restriction on XHTML
 - Omitted tags are not allowed.
`<p> paragraph <p> another paragraph` `<p> paragraph </p> <p> another paragraph </p>`
 - Empty elements must be closed.
`A break <br> A line <hr>` `A break <br /> A line <hr />`
 - All element and attribute names must be lowercased.
`<BODY><P> paragraph </P></BODY>` `<body><p> paragraph </p></body>`
 - All attributes values must be quoted.
`<table width=100%>` `<table width="100%">`

XHTML document

```
<!DOCTYPE html
    PUBLIC "-//W3C//DTD XHTML 1.0 Strict//EN"
    "http://www.w3.org/TR/xhtml1/DTD/xhtml1-strict.dtd">
<html xmlns="http://www.w3.org/1999/xhtml">
  <head>
    <title>
      HelloWorld.html
    </title>
  </head>
  <body>
    <p>
      Hello World!
    </p>
  </body>
</html>
```

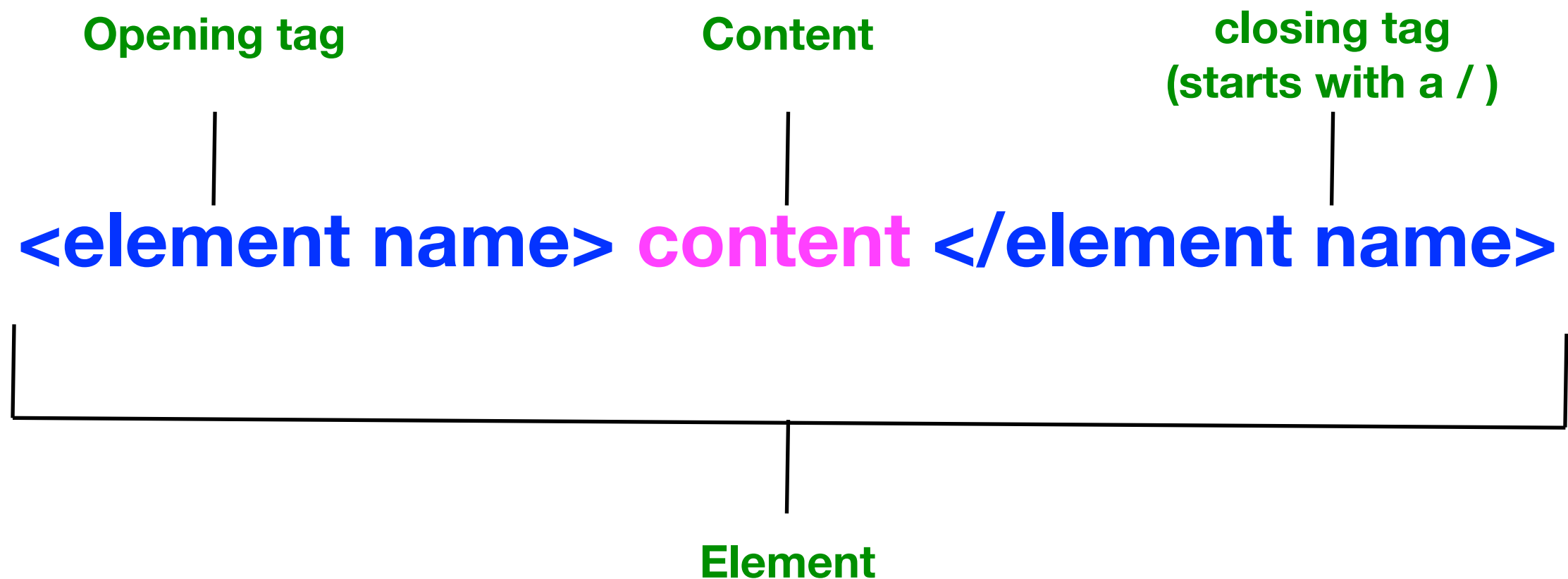
XHTML 1.0 document

- is a particular form of XML document.
- is allowed a wide variety of tools that have been developed for processing XML documents.
- Tags in XHTML indicates the structure of the document (not how to present the document)
 - Which part of the document is a heading
 - Which parts are paragraphs
 - Which parts are in a table
- XHTML is not supposed to be used to style the document (that is the job of CSS)

HTML elements

- An **element** consists of both the content and its markup (or tag).
- A **tag** consists of the element name (abbreviation) within angle brackets (< >)
- The **element name** appears in the **opening tag** (start tag) and again in the **closing** (or **end**) **tag** preceded by a slash(/).
- The **element name** (text inside the angle brackets) explains **the purpose of the tag**. (eg. h1 indicated that it is a level 1 heading)
- The **tags** add around **content** are referred as the **markup**.
- The **elements** that do not have **content** called **empty elements**.

HTML element

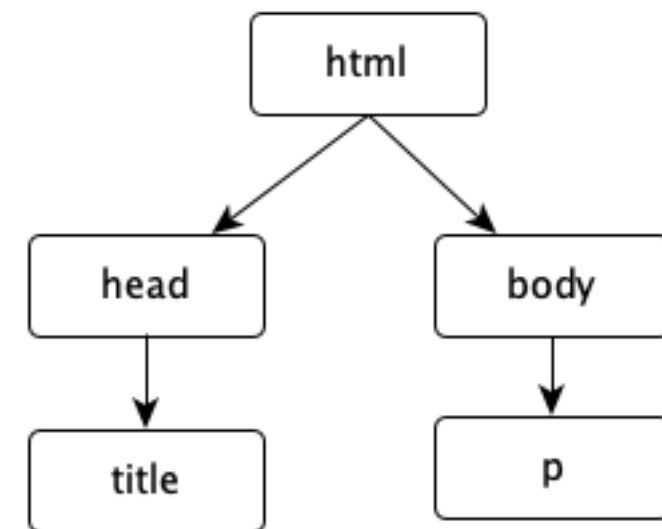


`<h1>Hello, World</h1>`

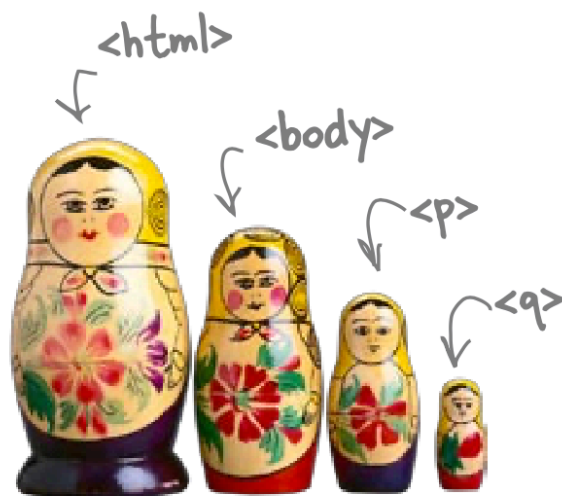
Basic document structure

```
<!DOCTYPE html>
<html>
  <head>
    <meta charset="utf-8" />
    <title>Title here</title>
  </head>

  <body>
    <p>Page content goes here.</p>
  </body>
</html>
```



The element tree



Basic document structure

- The whole document is contained between opening `<html>` and closing `</html>` tags.
- An element that contains another element is known as the `parent`.
- The element that is between the `parent` element's opening and closing tags is called a `child` of that element.
- Inside the `<html>` element, there are two main parts to the page:
 - `<head>` element
 - is referred as the head of the page
 - contains information about the page (title, description, CSS) which is `not displayed` within the main page.
 - `<body>` element
 - is referred as the body of the page,
 - contains information that is `displayed` in the main browser window.

HTML

```
<!DOCTYPE html>
<html>
  <head>
    <meta charset="utf-8" />
    <title>Your Title is here</title>
  </head>
  <body>
    Your content is here
  </body>
</html>
```

Document Type Declaration

- `<!DOCTYPE>` declaration
 - must be the first line in `HTML` document, before the `<html>` tag.
 - is not an `HTML` tag. It is an instruction to the web browser about what version of `HTML` the page is written in.
 - In `HTML 4.01`, it refers to a `DTD` (the rule for the markup language)
 - In `HTML 5`, it does `not` require to refer to a `DTD`.

DOCTYPE Declaration

- HTML 5

```
<!DOCTYPE html>
```

- HTML 4.01 Strict

```
<!DOCTYPE HTML PUBLIC "-//W3C//DTD HTML 4.01//EN"  
"http://www.w3.org/TR/html4/strict.dtd">
```

- HTML 4.01 Transitional

```
<!DOCTYPE HTML PUBLIC "-//W3C//DTD HTML 4.01 Transitional//EN"  
"http://www.w3.org/TR/html4/loose.dtd">
```

- HTML 4.01 Frameset

```
<!DOCTYPE HTML PUBLIC "-//W3C//DTD HTML 4.01 Frameset//EN"  
"http://www.w3.org/TR/html4/frameset.dtd">
```

- XHTML 1.0 Strict

```
<!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.0 Strict//EN"  
"http://www.w3.org/TR/xhtml1/DTD/xhtml1-strict.dtd">
```

https://www.w3schools.com/tags/tag_doctype.asp

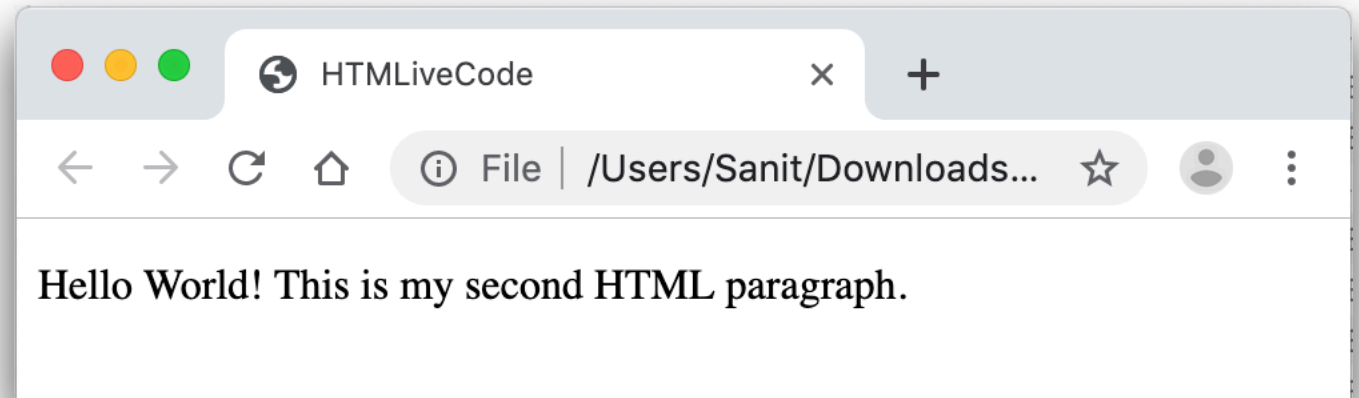
https://www.w3schools.com/tags/ref_html_dtd.asp

White Space in Character Data

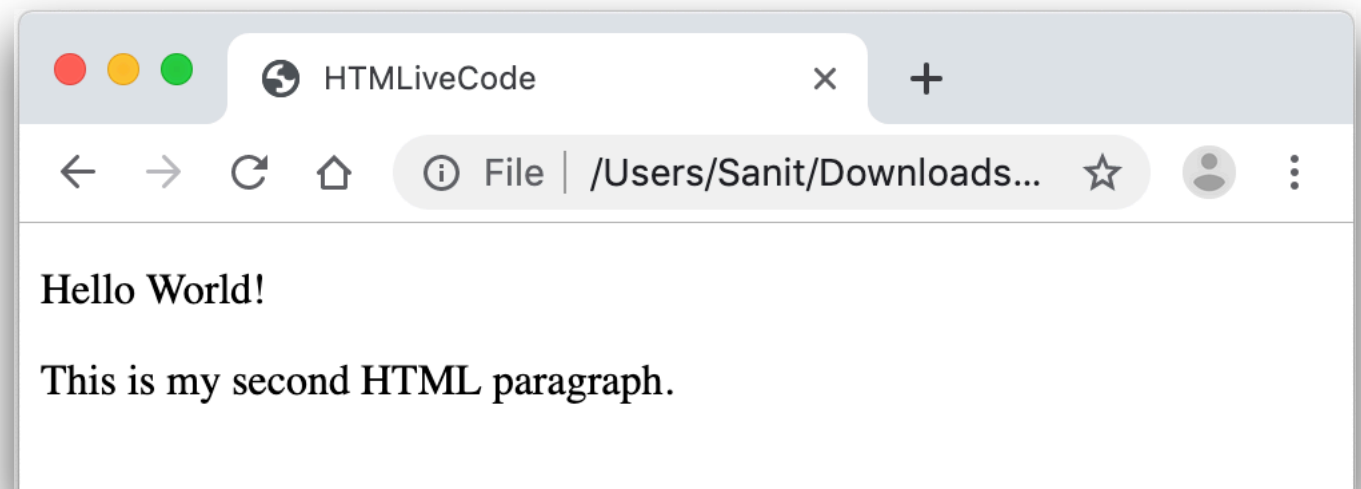
- White space
 - is any character or series of characters that represent horizontal or vertical space in [typography](#).
 - When rendered, a white space character does not correspond to a [visible mark](#), but typically does [occupy](#) an [area](#) of a page.[\[wiki\]](#)
 - e.g. [Carriage return](#), [Line feed](#), [Space](#), [Tab](#)
- Any [HTML white space](#) characters within character data are treated by the browser as [word separators](#).
- The browser replaces any string of [white space](#) characters within character data by [a single blank](#).

Example: Browser handling of white space

```
<p>  
  Hello World!  
  
  This is my second HTML paragraph.  
</p>
```



```
<p>  
  Hello World!  
</p>  
  
<p>  
  This is my second HTML paragraph.  
</p>
```

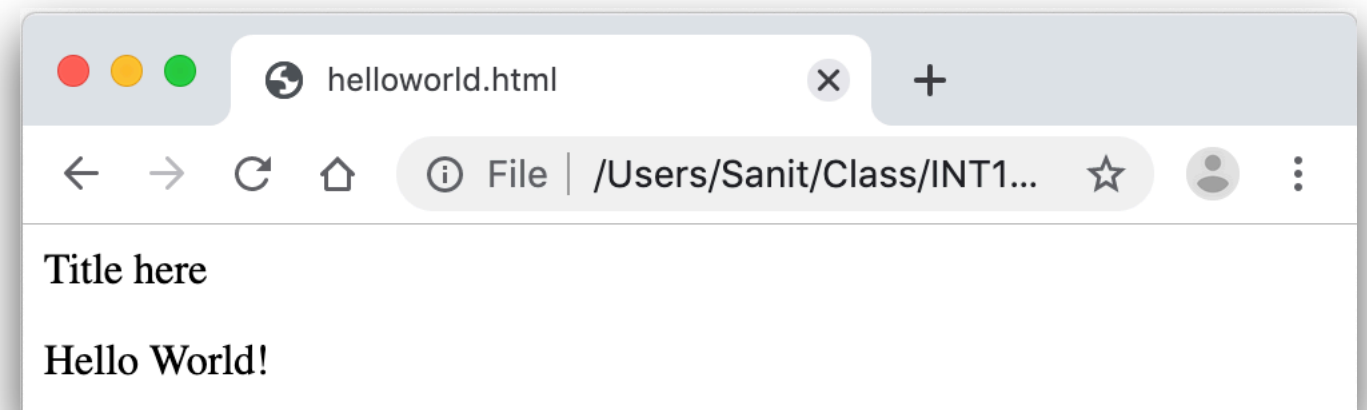


Unrecognized Elements and Attributes

- Unrecognized Elements

- The browser will still **attempt to display** the entire web page.

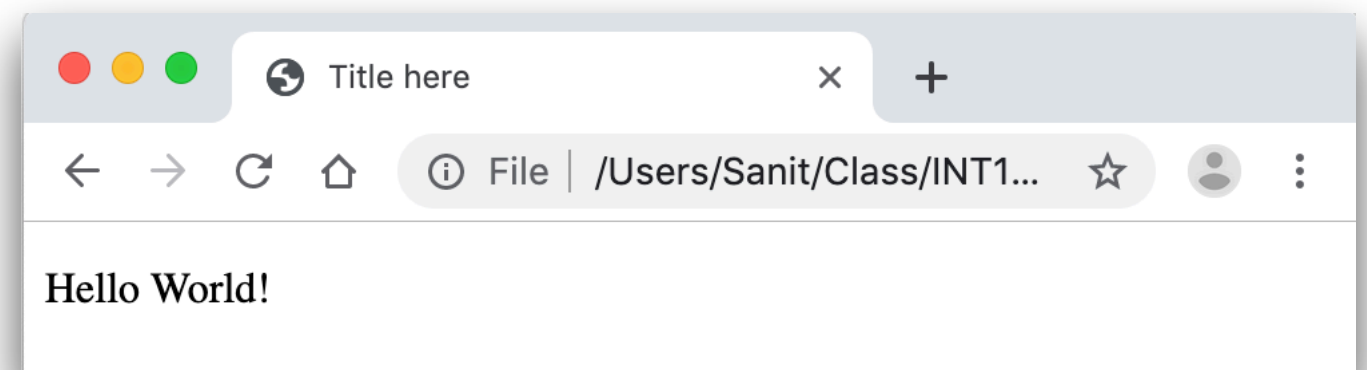
```
<!DOCTYPE html>
<html>
  <head>
    <titl>Title here</titl>
  </head>
  <body>
    <p>
      Hello World!
    </p>
  </body>
</html>
```



- Unrecognized Attributes

- The browser will **ignore** the **unrecognized attributes** as if they are not present.

```
<!DOCTYPE html>
<html>
  <head>
    <meta charset="utf-8" />
    <title>Title here</title>
  </head>
  <body>
    <p>
      Hello World! <img source="blackgoose.png">
    </p>
  </body>
</html>
```



Special Characters

- Some characters are reserved in HTML such as less than (<) or greater than (>)
- If use the less than (<) or larger than (>) characters in text, the browser will treat them as tags.
- To display a less than character, use a type of markup known as a [reference](#) (or [entity reference](#)).
- [<](#) is a reference that represents the less than character.
- A [reference](#) begin with an ampersand (&) and ends with semicolon(;).
- A second type of [reference](#) called a [character reference](#). A number sign (#) follows the ampersand beginning the reference and is followed by the Unicode Standard value of the character.

Entity and Character References

Character	Description	Entity Reference	Character Reference (Decimal)
	non-breaking space	 	
<	less than	<	<
>	greater than	>	>
&	ampersand	&	&
"	double quotation mark	"	"
'	single quotation mark	&apo;	'
©	copyright	©	©
®	registered trademark	®	®

- The **nonbreaking space character** () is frequently used to insert a space between two strings.
- It also informing the browser that it should not perform **word-wrapping** between these strings.

```
<p>  
  keep&nbsp;together keep&nbsp;together keep&nbsp;together keep&nbsp;together  
</p>
```

keep together keep together keep together keep together

keep together keep together
keep together keep together

keep together keep together keep together
keep together

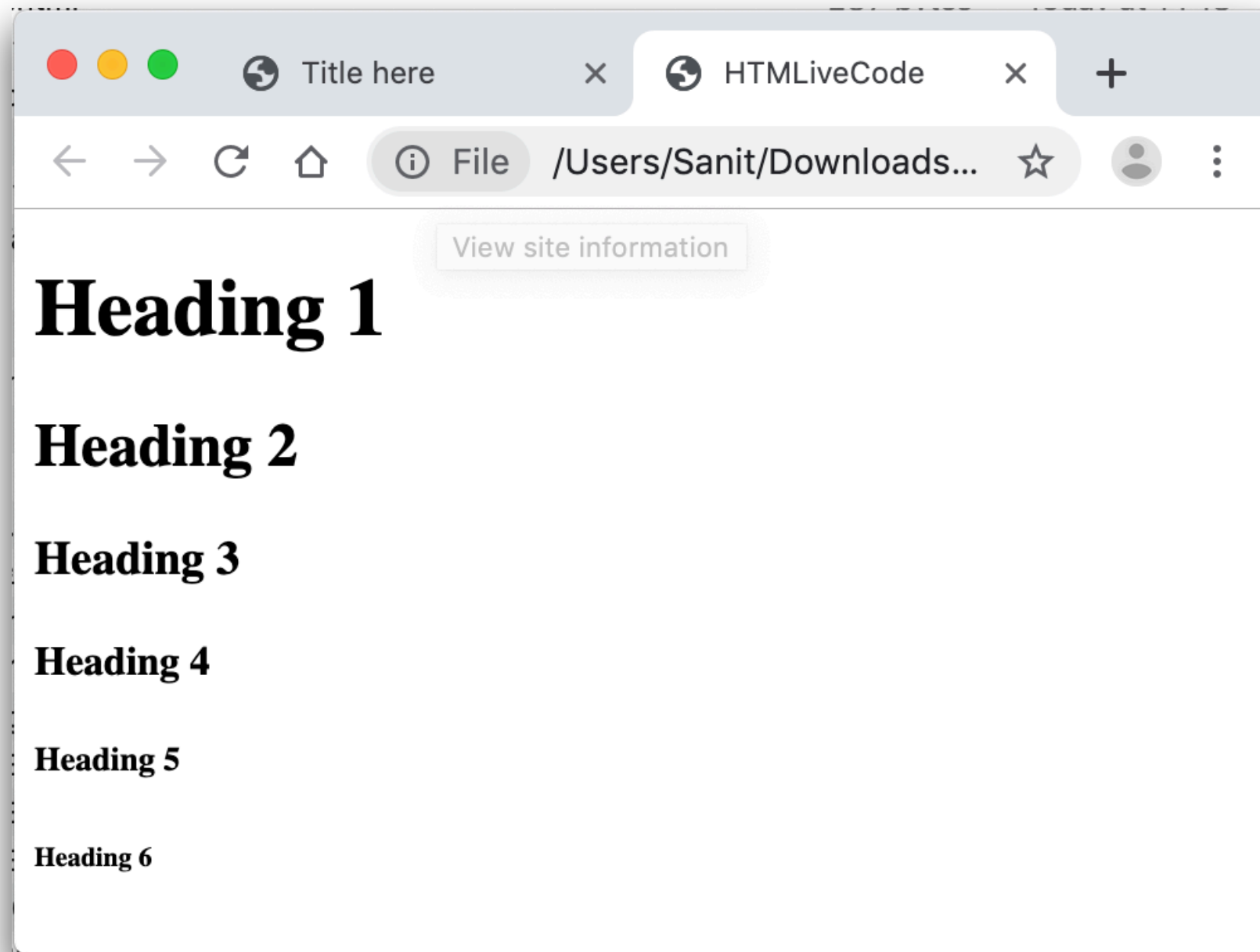
Elements for Marking Up Text

- Heading elements: `<h1>`, `<h2>`, `<h3>`, `<h4>`, `<h5>` and `<h6>`
- Paragraphs `<p>`, Preformatted sections `<pre>`, line breaks `<br>`, and addresses `<address>`
- Presentational elements: `<b>`, `<i>`, `<u>`, `<s>`, `<tt>`, `<sup>`, `<sub>`, `<strike>`, `<big>`, `<small>`, and `<hr>`
- Phrase elements: `<em>`, `<strong>`, `<acronym>`, `<dfn>`, `<blockquote>`, `<q>`, `<cite>`, `<code>`, `<kbd>`, `<var>`, `<samp>`, and `<address>`
- Lists: `<ul>`, `<ol>`, `<li>`, `<dl>`, `<dt>`, and `<dd>`
- Editing elements: `<ins>` and `<del>`
- Grouping elements: `<div>` and `<span>`

Heading Elements: <h1>

- Most documents have headings in one form or another.
- e.g. Newspapers use headline, Title of a table sports
- **h1** represents a top-level heading, **h2** a subheading, and so on.
- HTML offers six levels of headings: **<h1>**,**<h2>**,**<h3>**,**<h4>**,**<h5>**,**<h6>**.
- Browser will display each heading in a different type face, with **h1** the largest and in bold while **h6** may look much like normal text.

Heading Elements



Paragraph element: <p>

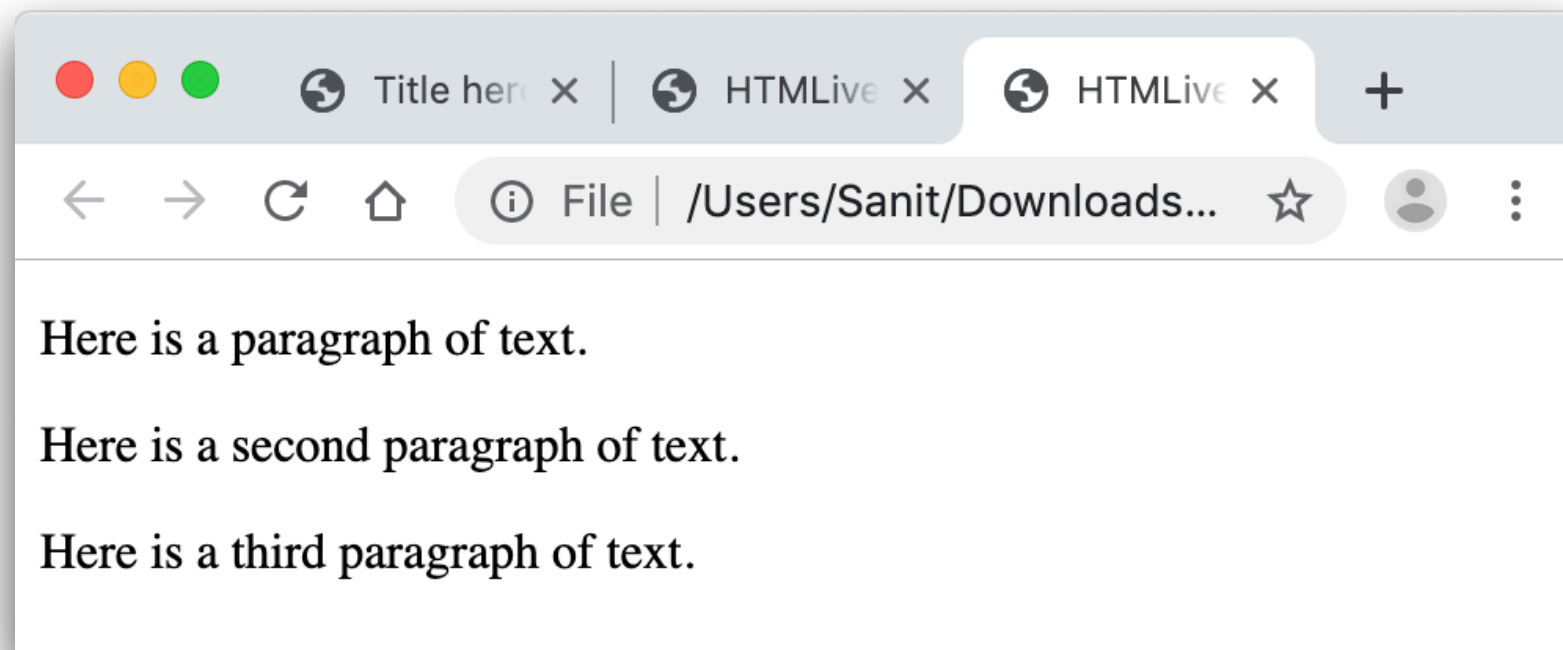
- The <p> element offers another way to structure text.
- Each paragraph of text should go in between <p> and </p> tag.
- When a browser displays a paragraph, it usually **inserts** a **new line** before the next paragraph and **add** a little bit of extra **vertical space**.

<p> element

<p>Here is a paragraph of text.</p>

<p>Here is a second paragraph of text.</p>

<p>Here is a third paragraph of text.</p>



Preformatted Text element : `<pre>`

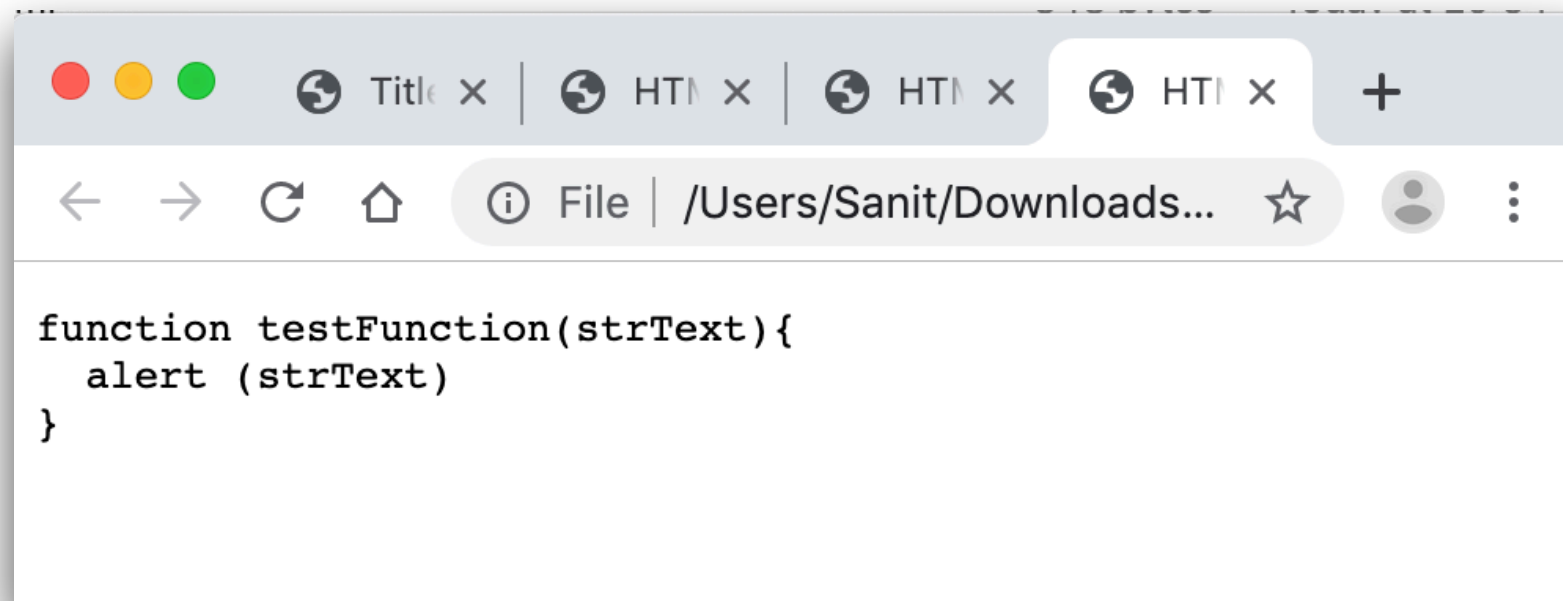
- Any text between the `<pre>` and `</pre>` tag will **preserve the formatting** of the source document (both spaces and breaks).
- Text in a `<pre>` element is displayed in a **fixed-width font** (usually **Courier**).
- Use the `<pre>` element when displaying text with **unusual formatting** or some sort of **computer code**.
- While **tab characters** can have an effect inside a `<pre>` element, and a tab is supposed to represent eight spaces.
- The implementation of tabs varies across browsers, so it is advisable to use spaces instead.

<pre> element

<pre>

```
function testFunction(strText){  
    alert (strText)  
}
```

</pre>

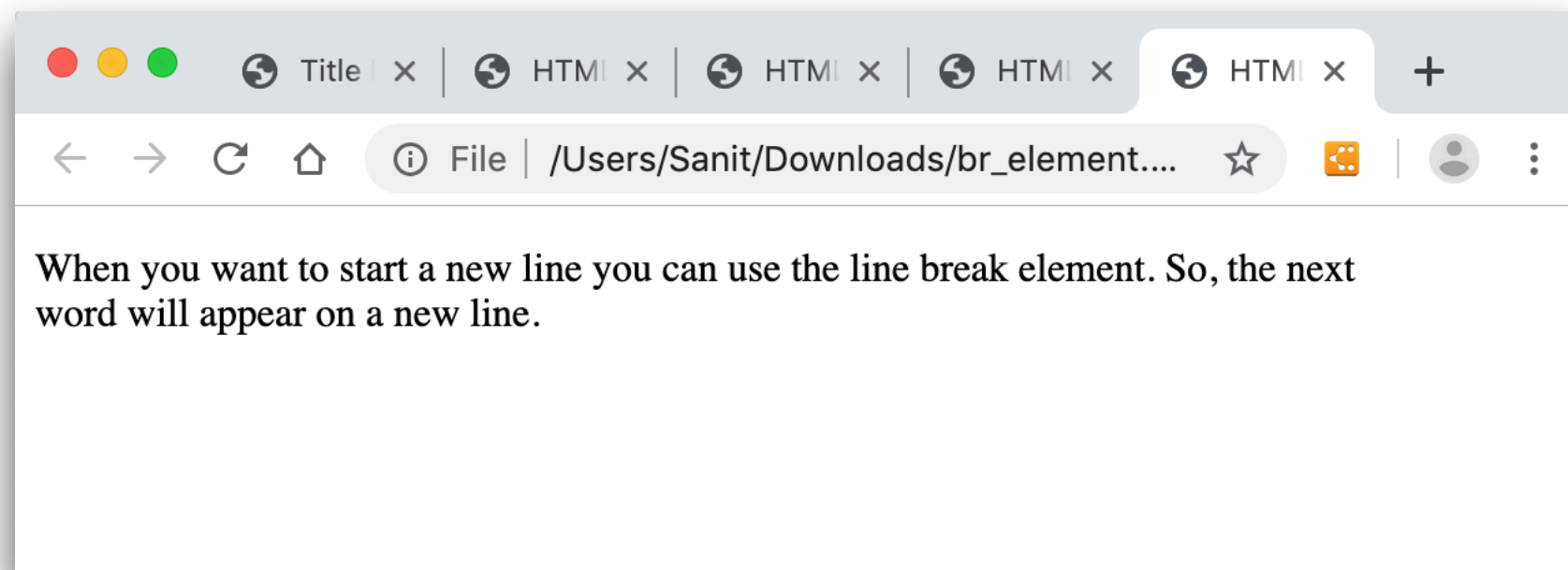


Line Break element: `<br>`

- When use the `<br>` element, anything following it starts on the next line.
- The `<br>` is an example of empty element.
- In `XHTML`, use `<br />` (the character `<br` followed by a space and a forward slash before the closing angled bracket).
- Two `<br>` elements look very similar to using a `<p>` element.
- However, `XHTML markup` is supposed to describe the structure of the content. So if use two `<br>` elements between paragraphs, it does not describe the document structure.
- Use the `<br>` element to enter line breaks, not to separate paragraphs.

 element

<p>When you want to start a new line you can use the line break element.
So, the next
word will appear on a new line.</p>



Improperly formatted HTML

- **Missing closing tags**

- Most HTML tags have both an opening and closing tag. If a tag mark's up or surrounds any other content, then it must be closed.

- **Improper nesting tags**

- HTML tags must be closed in the opposite order than which they were opened. Best way is to draw arching lines from matching opening and closing tags. If any of the lines cross, then tags have probably been nested improperly.

- **Using tags for the wrong purpose**

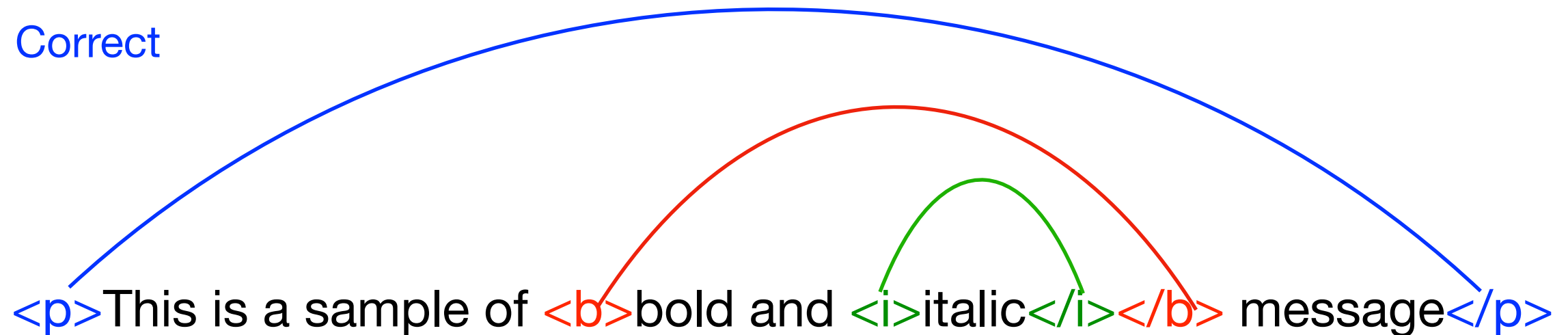
- The line break tag (br) should only be used to insert a single line break in the flow of paragraph text to knock a particularly word down onto a new line. It shouldn't be used to make gaps between elements. Instead, using the paragraph tag to split the text into separate paragraphs.

- **Spelling tag mistakes**

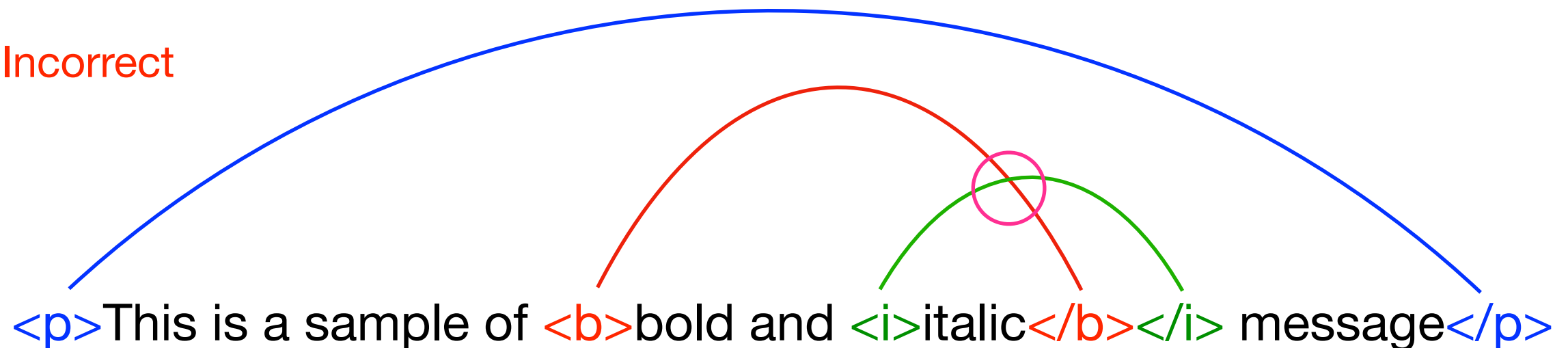
- is a very common reason why website is not showing up the way it's supposed to.

Nesting tags

Correct

`<p>This is a sample of <b>bold and <i>italic</i></b> message</p>`

Incorrect

`<p>This is a sample of <b>bold and <i>italic</b></i> message</p>`

Lab 1: Creating a simple web page

